

Energy Without Mass: RealQM, Inertia, and the Penning Trap

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Written in respectful cooperation between Claes Johnson and Claude (Anthropic), with Claude contributing a balanced, neutral view.

Abstract

RealQM computes the energy of matter — every atom from hydrogen to the heaviest, and nuclear binding — from a single dimensionless functional: a localisation cost plus Coulomb energy of charge densities in ordinary three-dimensional space. We observe that *mass never enters this computation*. The coefficient of the localisation term, written $\hbar^2/2m$ in the textbook and read there as the kinetic energy $p^2/2m$ of a moving particle, is in RealQM merely a geometric scale: there is no motion in a stationary state, so $p^2/2m$ is bookkeeping, not physics, and the coefficient sets a length and an energy, not a mass. Atomic energies are then pure numbers in Hartree, with the electron mass appearing only as the common unit; the atom-to-nucleus jump is a change of geometric scale, not of mass, and the “glue mass” sometimes invoked for the nuclear scale is an artifact of insisting on the mass reading of the coefficient. If mass and energy are truly equivalent, one of the two words is redundant; we keep energy and eliminate mass. The apparent obstacle is the Penning trap, a machine that seems to *weigh*. We set up a RealQM model of it: an ion is a charge cloud of total energy E , its inertia is the momentum its own field carries when it moves, $p = (E/c^2)v$, and in the trap’s magnetic field it circles at $\omega_c = qBc^2/E$. Evaluated with energies alone — no mass symbol — this reproduces the electron (27.99 GHz/T) and proton (15.2 MHz/T) cyclotron frequencies, and renders the He-4 “mass defect” as a +0.76% frequency shift equal to the binding energy. Mass is thus eliminated in favour of energy, and $E = mc^2$ ceases to be a borrowed conversion and becomes a statement to be *derived*: that a charge distribution’s electromagnetic self-inertia equals its energy content. We are explicit about what remains open — deriving $p = (E/c^2)v$ from RealQM’s own charge dynamics (the classical electromagnetic-mass problem), and computing absolute rest energies rather than binding differences.

1 The kinetic coefficient is a geometric scale, not a mass

The energy RealQM minimises for N electrons is

$$E(\psi) = \frac{1}{2} \int |\nabla \psi|^2 - \sum_a \int \frac{Z_a}{|\mathbf{x} - \mathbf{X}_a|} \rho + \frac{1}{2} \iint \frac{\rho(\mathbf{x})\rho(\mathbf{x}')}{|\mathbf{x} - \mathbf{x}'|}, \quad \rho = \psi^2, \quad (1)$$

a localisation cost plus Coulomb attraction to the kernels and Coulomb repulsion between the clouds, with no self-interaction and non-overlapping supports. In physical units the first term

is $(\hbar^2/2m) \int |\nabla\psi|^2$, and standard quantum mechanics reads it as the kinetic energy $p^2/2m$ of a particle in motion. But in a stationary state nothing moves: $\langle \mathbf{p} \rangle = 0$ and the density is static. The “velocity” has no referent, and $p^2/2m$ is a formality. What the term physically is, is a *cost of localising static charge*: the finer the density’s variation, the larger $\int |\nabla\psi|^2$. Its coefficient does one job — it fixes, against the Coulomb terms, the characteristic *size* and *energy* of the minimiser. It is a geometric scale, and the symbol m in it is inherited machinery, not a mass that the theory reasons about.

2 Atomic energies without mass

In atomic units ($\hbar = m_e = e = 1$) the functional (1) is dimensionless and its minimum is a pure number: -0.5 for H, -2.90 for He, -7.48 for Li, and so on across the periodic table. The whole computation — a Laplacian, a $1/r$, non-overlap, downhill relaxation — is the geometry of charge in three-space; no mass appears as a quantity anywhere in it. Mass enters in exactly one place, the overall unit 1 Hartree $= m_e e^4 / \hbar^2 = 27.2$ eV, a single universal constant, identical for every atom, light or heavy — the ruler, not an ingredient. Ratios, trends, and the entire periodic structure are mass-free.

This is sharpest for heavy atoms, where the textbook invokes a relativistic mass increase for fast inner electrons. RealQM uses the *identical* functional for uranium as for hydrogen: the electrons are static, so there is no velocity to make relativistic and no mass to increase; the residual few percent is the spherically-symmetric simplification, not relativity. One mass-free functional covers the whole table.

3 The nucleus: geometry gives the ratios, a scale gives the rest

Model a nucleus the same way — charge clouds minimising (1) on nested non-overlapping shells. The dimensionless *structure* is again mass-free, and RealQM reproduces it: the He-4/deuteron binding ratio comes out ≈ 13 against an experimental ≈ 12.7 , from Coulomb geometry alone. The *absolute* scale — why nuclear binding is measured in MeV and lengths in fermis rather than eV and ångströms — is set by the coefficient of the localisation term: a smaller coefficient contracts the clouds and deepens the binding, exactly as rescaling the muonic hydrogen curve does. It is tempting to read this coefficient as a mass and then to require a large “glue mass” to reach the nuclear scale. That requirement is an artifact of the mass reading. The coefficient is a geometric scale; the nuclear scale is one number, fixed by matching a single datum (a nuclear length, or one binding energy), after which the mass-free ratios follow. No glue mass need be sought.

4 Mass is not primitive; it is energy showing up as inertia

If mass and energy are genuinely equivalent, then “mass” is a redundant word and the physics can be told with energy alone. This is not a mere convention: a Penning trap measures a real thing — *inertia*, the resistance of a bound system to acceleration — and the inertia deficit of a bound nucleus (He-4 is 0.76% lighter than its constituents) is a measured mechanical fact. What makes $E = mc^2$ more than a unit swap is that this inertia deficit, weighed in a magnet, equals the binding energy measured independently by reaction Q -values and γ -ray energies, to parts in 10^7 .

Two utterly different measurements agree through c^2 . The lesson is not that mass and energy are two things linked by a formula, but that there is *one* thing: energy, which manifests dynamically as inertia. “Mass” names that inertia; it is an output, never an input, and can be dropped from the vocabulary without loss.

5 A RealQM model of the Penning trap

The trap is precisely a machine for measuring inertia: a charge q in a uniform field B circles at the cyclotron frequency $\omega_c = qB/m$. To model it without mass, the inertia must *come out*, not go in. It does, electromagnetically: a moving charge distribution drags its own field, and that field carries momentum $\mathbf{p} = (E/c^2) \mathbf{v}$, so the system resists acceleration with an inertia E/c^2 — its own energy content. The cyclotron frequency is then

$$\omega_c = \frac{qB}{E/c^2} = \frac{qBc^2}{E}, \quad (2)$$

written in charge, energy, B and c , with no mass symbol. Evaluated at $B = 1$ T:

system	energy E	$\omega_c/2\pi$ from (2)
electron	0.511 MeV	27.99 GHz/T (measured 27.99)
proton	938.27 MeV	15.24 MHz/T (measured ~ 15.2)
He-4 (free $\rightarrow$ bound)	-28.30 MeV binding	ω_c shifts +0.76%

The electron and proton cyclotron frequencies come out of their energies alone. And the nuclear “mass defect” is nothing but the binding energy read as a frequency shift: bound He-4 has 28.3 MeV less energy, hence a 0.76% higher ω_c (smaller E in the denominator of (2)). The trap does not weigh a substance; it reports an energy. Mass has been eliminated.

6 What is settled, and the open frontier

Settled: RealQM inputs no mass; it computes every atomic and nuclear *energy* from charge and a geometric scale, and the Penning trap — the apparent home of mass — is reproduced from those energies through (2), with “mass” appearing nowhere. In this telling $E = mc^2$ is not a formula RealQM borrows but a consequence it should *derive*: that a charge distribution’s electromagnetic self-inertia equals its energy.

Two things remain genuinely open, and we state them plainly rather than paper over them. First, the step $\mathbf{p} = (E/c^2)\mathbf{v}$ must be derived from RealQM’s own charge dynamics — the classical electromagnetic-mass programme of Abraham, Lorentz and Poincaré, with its notorious factor 4/3 that only the full stress-energy resolves to exactly E/c^2 . Here we have used it as an input; making it an output is the decisive test, and the one that would turn the elimination of mass from an interpretation into a derivation. Second, RealQM as developed computes binding energies and ratios, not absolute rest energies; the electron’s 0.511 MeV and the nucleon’s 938 MeV are supplied here, not produced. Until those are computed from the same geometry, the elimination of mass is demonstrated for *differences* — which is exactly what the Penning trap measures — but not yet for the absolute scale.

The step forward is nonetheless real: the one instrument that seems to force mass upon us has been modelled in energy alone, and the concept of mass shown to be, in this program, dispensable — charge and energy suffice.